

Osmania University

MID-TERM EXAMINATION ? 2026

Object Oriented Programming with Java ? Mid-Term Examination, 2026

Subject Code: CS303 Semester: 3 Regulation: R18

Time: 90 Minutes Max. Marks: 30

.....
Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
.....

SECTION A (Unit 1: Classes and Objects)

1. Define Classes and Objects. Explain its significance in Object Oriented Programming with Java with a suitable example. [12 Marks]
2. Compare and contrast the key concepts involved in Classes and Objects. [8 Marks]
3. Discuss the engineering applications of Classes and Objects. [12 Marks]

SECTION B (Unit 2: Inheritance)

4. Derive the expression related to Inheritance and explain each term. [12 Marks]
5. How is Inheritance implemented? Give an algorithm / procedure and explain. [12 Marks]
6. Solve a problem related to Inheritance. Show all the steps clearly. [8 Marks]

SECTION C (Unit 3: Polymorphism and Abstraction)

7. How is Polymorphism and Abstraction implemented? Give an algorithm / procedure and explain. [12 Marks]
8. Solve a problem related to Polymorphism and Abstraction. Show all the steps clearly. [8 Marks]
9. List the advantages and limitations of Polymorphism and Abstraction. Explain any two in detail. [12 Marks]

SECTION D (Unit 4: Packages and Interfaces)

10. Write short notes on Packages and Interfaces. [12 Marks]
11. Describe the working principle of Packages and Interfaces with necessary equations. [12 Marks]
12. Discuss the engineering applications of Packages and Interfaces. [12 Marks]

SECTION E (Unit 5: Exception Handling and Multithreading)

13. Discuss the engineering applications of Exception Handling and Multithreading. [12 Marks]
14. How is Exception Handling and Multithreading implemented? Give an algorithm / procedure and explain. [12 Marks]
15. List the advantages and limitations of Exception Handling and Multithreading. Explain any two in detail. [12 Marks]

*** End of Question Paper ***